

Revision of mathematics, told as business decisions

MBA03 Research & Quantitative Business Methods — Week 1

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Thursday 24 September 2026

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Where the first five weeks go

Wk	Date	What you will be able to do by hand
1	24 Sep	equations, systems, quadratics, logs
2	1 Oct	read a function; limits; the derivative
3	8 Oct	study a function; optimise; integrate
4	15 Oct	interest, discounting, NPV
5	22 Oct	bonds, duration, convexity; mock exam

Two things count.

Midterm (25%): a report to a Board built on the study of a cost, a revenue and a profit function.
Due 1 Nov.

Final exam (75%): 3 hours, paper, calculator. One of five questions is pure calculus: integrals, limits, recovering a function from its derivative.

Every technique in this room is here because one of those two asks for it. I will tell you which one, every time.

The shape of an exam item

Same form as the final, different numbers:

$$\int \frac{x^5 - 3x^3 + 2}{x^2} dx$$

$$\lim_{x \rightarrow 1^+} \frac{(x-1)^2}{x-x^2}$$

$$f'(x) = 6x^2 - 2x, \quad f(1) = 4 \Rightarrow f(x) = ?$$

Today we do none of this. Today we make sure that when we get there, the *algebra* underneath — dividing term by term, factoring, solving for an unknown, handling a log — is not what slows you down.

Everything today is a question the founder of a small firm actually has to answer.

The case: Arno Bikes

e-bike assembler, Florence

Fixed cost €60,000 per month (rent, salaries, insurance)

Variable cost €400 per bike (parts, packaging)

List price €1,000 per bike

Four questions from the founder, one per block:

- A.** How many bikes before I stop losing money? **B.** How many bikes should I make?
C. How long until revenue doubles? **D.** Is this battery cell within spec?

A. Launch or not? — linear equations and systems

How many bikes a month before Arno Bikes stops losing money?

Bet before we compute

Write a number. Do not compute yet.

Two facts, written as two functions of the monthly quantity q :

$$C(q) = 60,000 + 400q \qquad R(q) = 1,000q$$

How many bikes a month before Arno Bikes stops losing money?

Bet before we compute

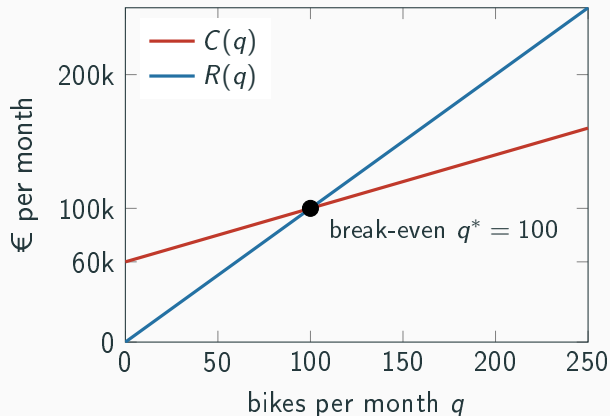
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Two facts, written as two functions of the monthly quantity q :

$$C(q) = 60,000 + 400q \qquad R(q) = 1,000q$$

Break-even is where they meet: $R(q) = C(q)$.

Two lines and a crossing



$$1,000 q = 60,000 + 400 q$$

$$600 q = 60,000$$

$$q^* = \frac{60,000}{600} = 100$$

Reading. €600 is the *contribution* of each bike. Break-even is fixed cost $\div$ contribution.

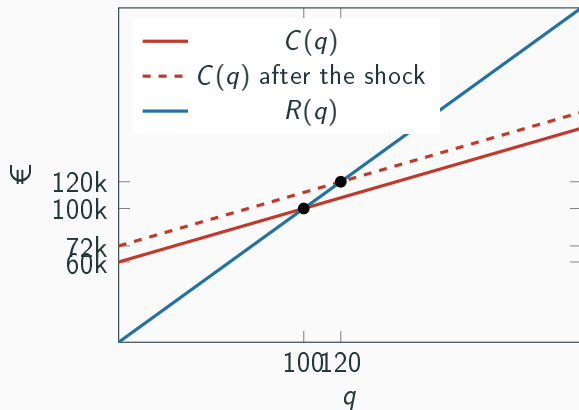
LAB `session.ipynb` §A

Move the price. Then raise the rent.

1. Bet: what price makes break-even 50 bikes? Then slide.
2. The landlord raises fixed cost to €72,000. Before running: what happens to the cost line, and to the crossing?

Watch the *slope* and the *intercept* separately. Only one of them moved.

A cost shock is a parallel shift — remember this for the midterm



Same slope (€400 per bike), higher intercept.

New break-even:

$$1,000q = 72,000 + 400q \Rightarrow q^* = 120.$$

The midterm asks how a tariff on an input changes the cost and profit *schedules*. That is this picture with different names: a per-unit tariff changes the slope, a lump-sum cost changes the intercept.

The whole market: two facts, two unknowns

Buyers in Tuscany: $Q_d = 1200 - 2p$. Sellers: $Q_s = -300 + 3p$.

Bet before we compute

What price clears the market?

Substitution

$$1200 - 2p = -300 + 3p$$

$$1500 = 5p$$

$$p^* = 300, Q^* = 600$$

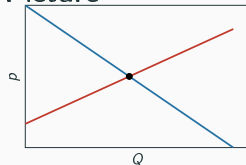
Elimination

$$Q + 2p = 1200$$

$$Q - 3p = -300$$

$$\text{subtract: } 5p = 1500$$

Picture



Board sentence

One equation is one fact about the business. Two unknowns need two facts. Write the facts as equations before touching a number.

PAPER drills.md

D1, D2 (core) — D3 (stretch)

10 minutes. Show all work, exam style. Tutor circulates.

B. How many bikes? — quadratics

Price responds to volume

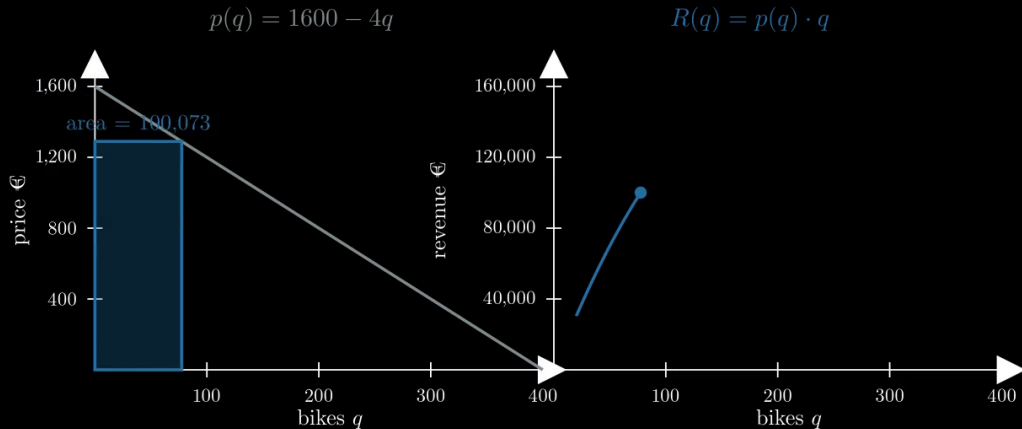
The founder learns: at €1,000 she sells 150 bikes; every €4 cut sells one more.

$$p(q) = 1600 - 4q$$

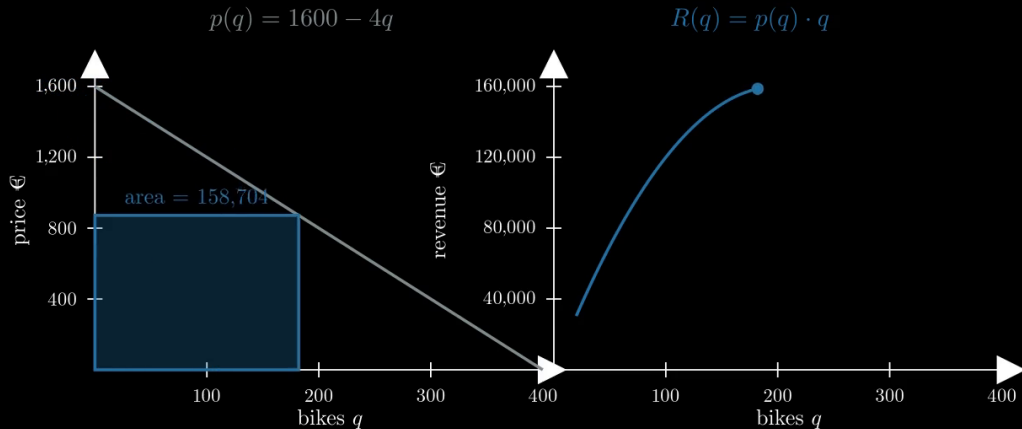
Revenue is **price** $\times$ **quantity**: a rectangle whose sides pull in opposite directions.

Bet before we compute

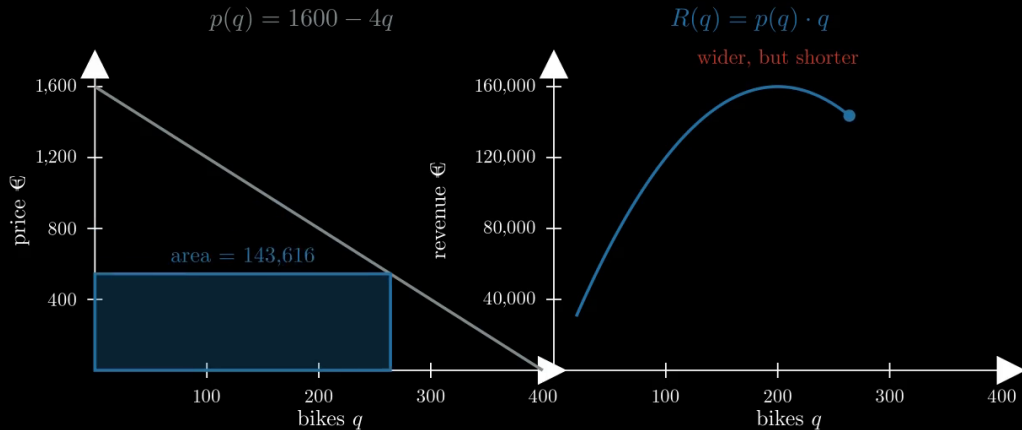
Does selling more *always* bring more revenue? Yes / no, and roughly where it stops.



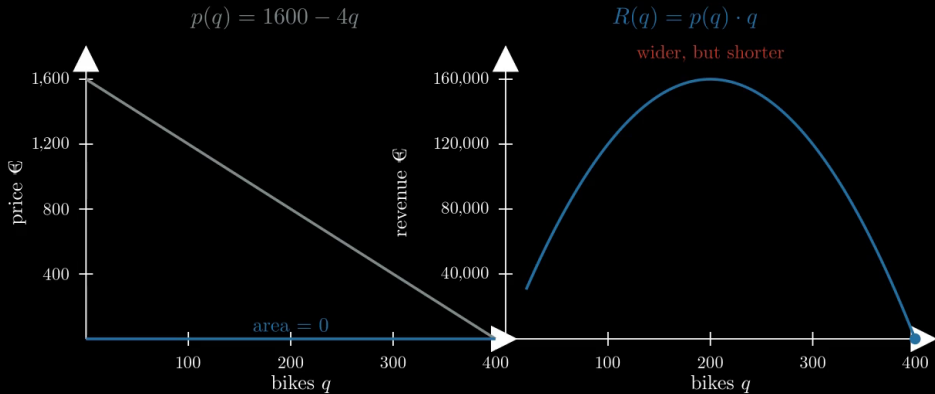
► play w01_parabola.mp4 (frames rendered with manim from `anim/src/week01_scenes.py`)



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Revenue is the area of a rectangle whose sides pull in opposite directions.

► play w01_parabola.mp4 (frames rendered with manim from anim/src/week01_scenes.py)

$$R(q) = (1600 - 4q)q = 1600q - 4q^2$$

$$\pi(q) = R(q) - C(q) = -4q^2 + 1200q - 60,000$$

$$R(q) = (1600 - 4q)q = 1600q - 4q^2$$

$$\pi(q) = R(q) - C(q) = -4q^2 + 1200q - 60,000$$

Where is profit zero? First move, always: divide out the common factor.

$$q^2 - 300q + 15,000 = 0$$

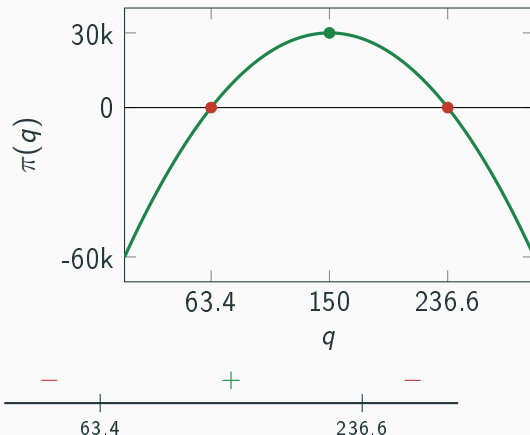
$$q = \frac{300 \pm \sqrt{300^2 - 4 \cdot 15,000}}{2} = \frac{300 \pm \sqrt{30,000}}{2} \approx 63.4, 236.6$$

The discriminant decides before you compute

$b^2 - 4ac$	roots	in business
> 0	two	a profitable window
$= 0$	one	a single break-even
< 0	none	no viable scale

Where is profit positive? A downward parabola ($a < 0$) is positive *between* its roots:

$$63.4 < q < 236.6$$



The sign line. In the midterm it is called “the sign of the profit function”.

LAB `session.ipynb §B`

Push the fixed cost up. Where do the two break-evens merge?

1. Bet on the fixed cost at which the roots become one. (Hint: what is the profit at the vertex now?)
2. Then push further. What does “no roots” look like, and what does it mean for the founder?

The top of the hill, for now

Vertex of $\pi(q) = -4q^2 + 1200q - 60,000$:

$$q_v = -\frac{b}{2a} = -\frac{1200}{-8} = 150 \quad \pi(150) = 30,000$$

This formula works for parabolas only.

Next week you get a tool that finds the top of the hill for *any* curve — a log revenue, a cubic cost, anything. It is called the derivative, and it is the reason this course exists.

Board sentence

A quadratic has two roots, one, or none, and the discriminant tells you which before you compute. In business: a profitable window, a single break-even, no viable scale.

PAPER drills.md

D4, D5 (core) — D6 (stretch: factor and cancel)

10 minutes. Show all work, exam style. Tutor circulates.

C. How long to double? — exponentials and logarithms

Revenue €1.2M, growing 12% a year

$$R(t) = 1.2 \cdot 1.12^t$$

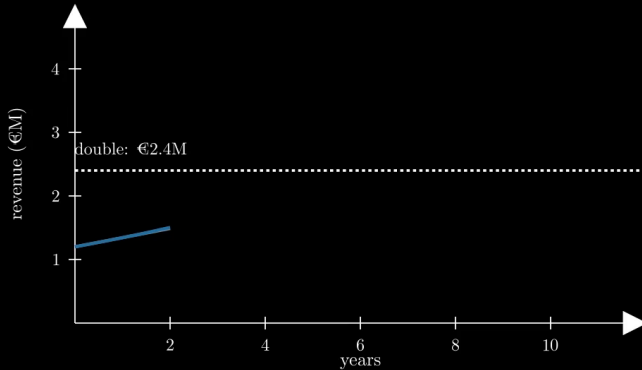
Linear growth adds the same **amount** each year. Exponential growth adds the same **percentage**.

Bet before we compute

How many years to double? Write a number.

linear: +€144k every year

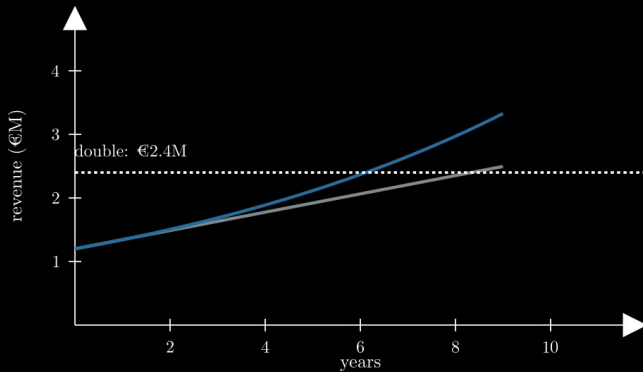
exponential: $\times 1.12$ every year



► play w01_growth.mp4 (frames rendered with manim from `anim/src/week01_scenes.py`)

linear: +€144k every year

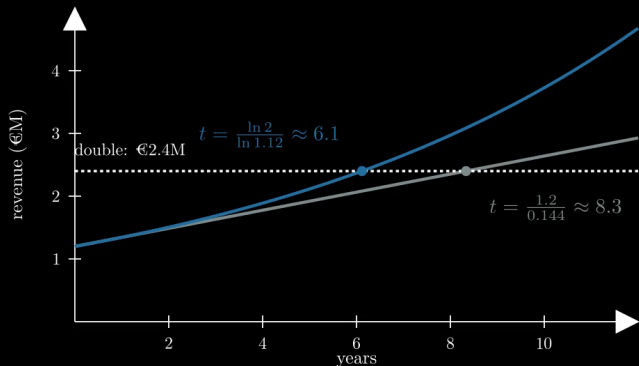
exponential: $\times 1.12$ every year



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linear: +€144k every year

exponential: $\times 1.12$ every year



Linear adds the same amount; exponential adds the same percentage.

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The unknown is in the exponent

$$1.2 \cdot 1.12^t = 2.4 \implies 1.12^t = 2$$

No algebra so far reaches t . **That is what a logarithm is for:** it answers “to what power?”

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$$\ln(1.12^t) = \ln 2 \implies t \ln 1.12 = \ln 2 \implies t = \frac{0.6931}{0.1133} \approx 6.1 \text{ years}$$

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rule	reading
$\ln(ab) = \ln a + \ln b$	growth factors multiply, their logs add
$\ln(a/b) = \ln a - \ln b$	a ratio of revenues becomes a difference
$\ln a^k = k \ln a$	k years of the same growth: the exponent comes down
$\log_b x = \ln x / \ln b$	any base through the calculator's $\ln$

The sanity check you will use in the exam

$$t_{\text{double}} \approx \frac{70}{g\%} \qquad \frac{70}{12} \approx 5.8$$

Why it works: $\ln 2 \approx 0.69$ and $\ln(1 + g) \approx g$ when g is small, so $t = \ln 2 / \ln(1 + g) \approx 0.69/g$.

The number $e \approx 2.718$ appears today only as “the base your calculator’s $\ln$ uses”. In week 4 it becomes continuous compounding and earns its place.

Board sentence

Whenever the unknown is in the exponent, take logs of both sides. Whenever you see a log, remember it is a question: to what power?

PAPER drills.md

D7, D8 (core) — D9 (stretch: $5e^{0.03t} = 8$)

12 minutes. Show all work, exam style. Tutor circulates.

D. Within tolerance — absolute
value

Is this cell within spec?

Battery cells must be $500 \text{ Wh} \pm 15$. A supplier ships one at 483 Wh.

Bet before we compute

In or out? Then write the rule as one inequality.

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Bet before we compute

In or out? Then write the rule as one inequality.

$$|c - 500| \leq 15 \iff 485 \leq c \leq 515 \iff c \in [485, 515]$$



$|x - a|$ is the **distance** from a . “ $\leq b$ ” is the closed interval $[a - b, a + b]$; “ $> b$ ” is the two tails $x < a - b$ or $x > a + b$.

Square bracket: endpoint included. Round bracket: excluded, or $\pm\infty$. The exam asks for domains written exactly like this.

PAPER drills.md

D10, D11 (core) — interval notation

8 minutes. Show all work, exam style. Tutor circulates.

Board pitch and close

Sixty seconds to the Board

One of you stands up. The other is the Board.

Tell them, with numbers and no formulas:

- where Arno Bikes breaks even at today's price;
- between which volumes it makes money once price responds to volume, and how fragile that window is;
- how long until revenue doubles at 12%.

The Board asks one question.

This is a rehearsal of the midterm's oral.

Take home

1. Write the facts as equations before touching numbers.
2. Discriminant first, then roots, then the sign line.
3. Unknown in the exponent $\rightarrow$ take logs of both sides.

Homework 1 (`homework.md`), due Wednesday 30 Sep 23:59 on Moodle: eight drills on paper + a 150–250-word memo to the Board on the profitable window.

Write your own three take-home lines now, before you leave.